

ACTIVITY 14 — LEARNER WORKSHEET**Risk Register, RBS, EMV and Decision Tree**

TYPE	DURATION	MAPS TO	TOPIC
Workshop	90 minutes	LO4	3 - Plan the Project

What you are doing

Contoso Learning Pte Ltd is a training provider with 14,000 active learners. Its course registration portal is 9 years old. Registration takes a median of 11 minutes and 34% of attempts are abandoned. Learner communications consume 24 staff-hours per week across 3 administrators. 62% of learner traffic is mobile, against a portal that is not responsive. Legacy consent screens do not meet current PDPA requirements. The project modernises course registration and learner communications.

What you will produce: risk register, rbs, pi matrix — submitted to artifacts/.

What you need before you start

- Lab 02 PESTLE/TECOP factors
- Lab 06 assumption log A-01 to A-07
- Lab 13 contingency reserve
- Lab 04 compliance threats

Your steps

1. Build the risk breakdown structure. Create artifacts/14-rbs.md. The RBS is a hierarchy of risk SOURCES. Its purpose is systematic identification: working through categories catches risks that free brainstorming misses, because brainstorming surfaces what is memorable rather than what is likely.
2. Trace risks from Lab 02 factors and Lab 06 assumptions. Risk identification is not invention. Most risks on a well-run project are already sitting in earlier artifacts, unlabelled.
3. Write risk statements in cause-event-effect form
4. Build the risk register. Create artifacts/14-risk-register.md. Probability is given as a decimal and as a 1-5 score. Impact is scored 1-5, with the cost impact in SGD and the schedule impact in days.
5. Plot the probability/impact matrix. Create artifacts/14-pi-matrix.md. Score = probability score x impact score, both on 1-5.
6. Apply all ten response strategies. A real Contoso example of each of the ten:
7. Compute expected monetary value. Sign convention: threat impacts are shown as positive cost figures and summed as exposure; opportunity impacts are shown as negative because they reduce exposure.
8. Build and solve a decision tree. Create artifacts/14-emv-analysis.md. A decision tree evaluates a choice under uncertainty by computing the EMV of each branch and selecting the best.
9. Qualitative versus quantitative, and when a risk becomes an issue
10. Answer the exam-style scenarios

Your working space

Complete every field. Keep this worksheet — it is your reference for the rest of the course.

Step 1 — Build the risk breakdown structure

Step 2 — Trace risks from Lab 02 factors and Lab 06 assumptions

Step 3 — Write risk statements in cause-event-effect form

Step 4 — Build the risk register

Step 5 — Plot the probability/impact matrix

Step 6 — Apply all ten response strategies

Step 7 — Compute expected monetary value

Step 8 — Build and solve a decision tree

Step 9 — Qualitative versus quantitative, and when a risk becomes an issue

Step 10 — Answer the exam-style scenarios

Deliverable checklist

- 14-rbs.md - the risk breakdown structure to two levels across Technical, External, Organisational and Project Management, with the risk count by category and a comment on concentration.
- The traceability showing which Lab 02 factors and which Lab 06 assumptions generated which risks, with at least one conversion worked in full.
- 14-risk-register.md - 16 to 18 risks including at least two opportunities, each with a cause-event-effect statement, probability as a decimal and a 1-5 score, impact in SGD and days, Pxl score, response strategy, owner, trigger, secondary risk and residual probability.
- 14-pi-matrix.md - the 5x5 grid with every risk plotted by ID, the red/amber/green zones defined, and the tie to the Lab 03 escalation threshold of 20 stated.
- A real Contoso example of each of the ten response strategies - five threat, five opportunity - including both active and passive acceptance.
- 14-emv-analysis.md - EMV per risk gross and residual, the totals, and the reconciliation to the SGD 26,000 contingency reserve with the variance stated.
- The fully worked decision tree with branch probabilities summing to 1.0 at each chance node, decision node costs included, both path EMVs computed, the recommendation stated, and a sensitivity analysis.
- The qualitative versus quantitative comparison and the risk-to-issue transition sequence.
- Written answers to the three scenarios.